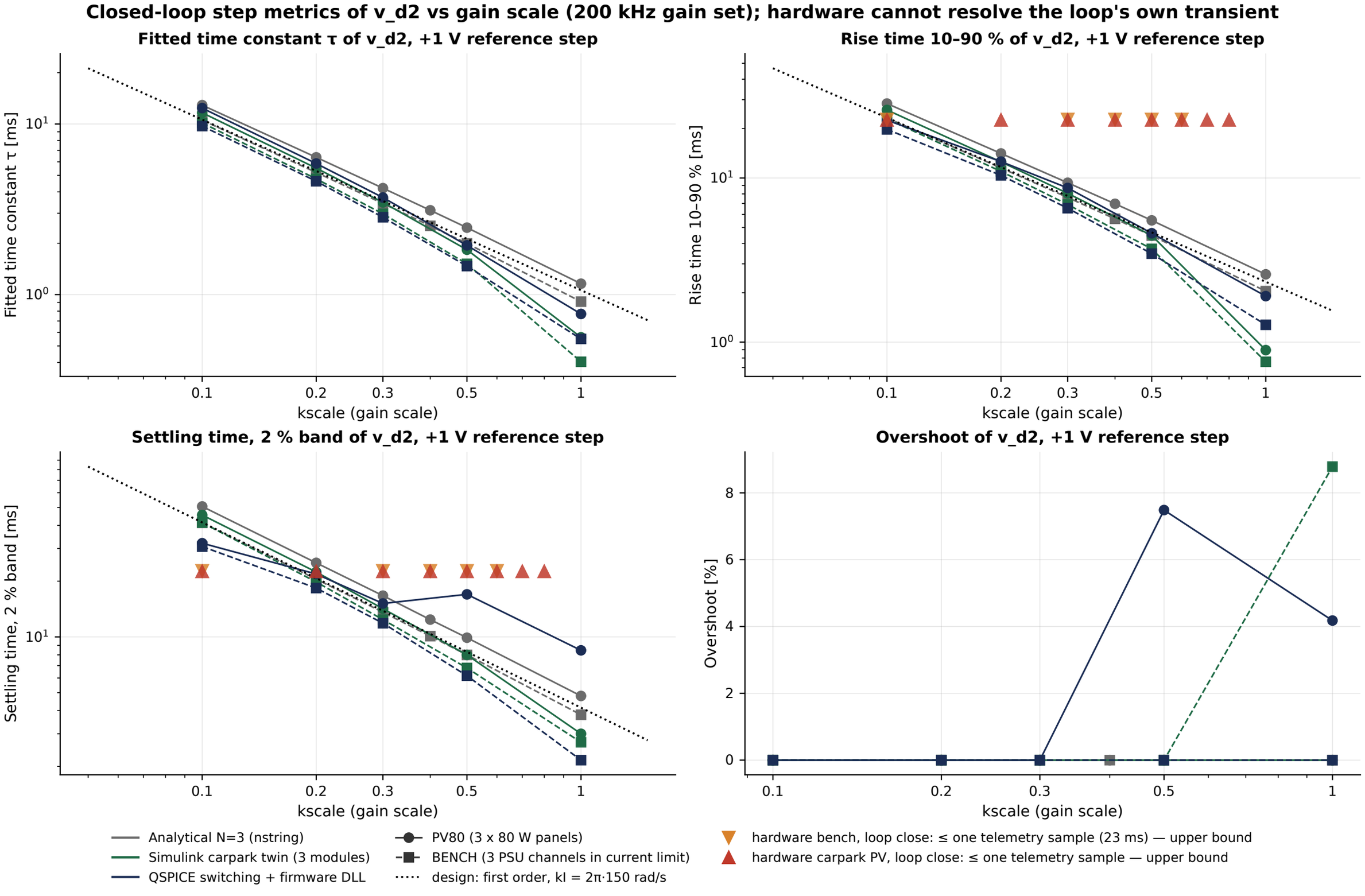
Closed-loop step metrics of v_d2 vs gain scale (200 kHz gain set); hardware cannot resolve the loop's own transient
Fitted time constant τ of v_d2, +1 V reference step
Fitted time constant τ [ms]
kscale (gain scale)
Rise time 10–90 % of v_d2, +1 V reference step
Rise time 10–90 % [ms]
kscale (gain scale)
Settling time, 2 % band of v_d2, +1 V reference step
Settling time, 2 % band [ms]
kscale (gain scale)
Overshoot of v_d2, +1 V reference step
Overshoot [%]
kscale (gain scale)
Analytical N=3 (nstring)
Simulink carpark twin (3 modules)
QSPICE switching + firmware DLL
PV80 (3 x 80 W panels)
BENCH (3 PSU channels in current limit)
design: first order, kI = 2π·150 rad/s
hardware bench, loop close: ≤ one telemetry sample (23 ms) — upper bound
hardware carpark PV, loop close: ≤ one telemetry sample — upper bound